

Correlation between economic, health, and institutional context of applicants' country and INZ student visa approval decisions

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Abstract. This article examines country-level variation in Immigration New Zealand fee-paying student visa approval rates for the 2022–2025 period. The study combines publicly available INZ decision data with publicly available country-level reference variables from the World Bank, the World Health Organization, and UN M49 regional classifications. The analysis uses descriptive statistics, Pearson correlations, linear regression approximations, alternative regional specifications, residual diagnostics, and a weighted robustness check based on log-transformed application volume.

The results show that approval rates vary systematically across applicant nationalities and are associated with application volume, income classification, regional classification, and broader country-level development indicators. In the modelling table, several health, income, internet-use, mortality, and life-expectancy indicators are strongly correlated with approval rates, although these indicators also overlap substantially with one another. Among the tested regression specifications, the UN M49 regional model has the strongest fit, followed by the World Bank regional model, the WHO regional model, and the no-region model. A weighted UN M49 experiment gives similar overall fit but weakens several individual regional coefficients, supporting the model-fit conclusion while limiting interpretation of particular coefficients.

The findings should be interpreted as country-level descriptive evidence, not as causal evidence about individual visa decisions. The analysis is explicitly based on INZ's published visa instructions, which set out common formal requirements for student visa applicants, including evidence of enrolment, tuition arrangements, sufficient funds, bona fide temporary-entry intent, and relevant health and character requirements. The observed country-level differences are therefore not interpreted as evidence of officer behaviour, discriminatory intent, bias, or unequal treatment in individual application assessment. Rather, they are treated as aggregate descriptive outcomes that may reflect country-level differences in documentary, financial, institutional, and verification conditions under the same formal decision framework. The principal limitation is that the data are aggregated by country and do not include applicant-level records, programme choice, education provider information, documentation quality, or detailed processing factors. The study identifies a reproducible empirical structure and points to further research on other receiving countries, additional country-level variables, and formal inter-country education or migration relationships.

Keywords: student visas · Immigration New Zealand · country-level analysis · approval rates · regional classification

1 Introduction

Record number of higher education students highlights global need for recognition of qualifications [25]. This rapid expansion reflects a global broadening of higher education pathways, including online degrees, hybrid courses, and micro-credentials, complementing traditional models and reshaping how knowledge is delivered and assessed. Academic mobility is also on the rise, with 6.9 million students studying away from their home country, a number that has tripled since 2000.

TODO: State the empirical problem and research question early. Keep the framing narrow: the article studies observed variation in INZ fee-paying student visa approval rates by applicant nationality and country-level reference data.

Prior quantitative studies have combined international student mobility or visa data with country-level development indicators, especially World Bank variables. However, peer-reviewed studies combining student visa decision data with both World Bank development indicators and WHO health indicators appear limited, making the present country-level descriptive design a contribution to the literature.

Previous quantitative research on international student migration has used visa refusal or visa-policy data together with student mobility or enrolment outcomes, particularly in the United States and Canada. Chen examines U.S. F-1 visa restrictiveness and international student enrolment, while Gonnot studies Canadian student visa policy and community-college enrolment [6, 8]. Other work models international student mobility using origin- and destination-country characteristics, including World Bank development indicators. However, the literature contains fewer studies that directly analyse country-level student visa approval rates as the dependent variable and combine them with broader country-level health and development indicators. This article addresses that gap using INZ fee-paying student visa decision data joined with World Bank, WHO, and UN M49 country-level variables.

No works about NZ student visa.

Why NZ in good as example? – Small, isolated, with big revenue from education exporting.

Add share of international students in NZ of all international students in the world.

Hard to apply. Not cheap. No noise applications.

No explicit unequity in rules.

2 Literature Review

“Some are born to Sweet Delight, Some are born to Endless Night” [4].

International education is often presented as an open global market in which students choose programmes and institutions compete for talent. This framing hides a basic constraint: students cannot simply choose to study abroad unless a state permits them to enter and remain. Cross-border education is therefore not only an academic or economic process; it is also a migration process governed by visa systems. Sassen notes that international law does not provide a general right to enter foreign territory [22]. Shachar similarly emphasises that the practical value of citizenship is closely tied to mobility rights attached to passports [23]. Visa systems therefore sit at the boundary between education markets and state control over entry.

This matters because mobility is unevenly distributed. Neumayer argues that states use visa restrictions to manage a trade-off between facilitating entry for some passport holders and deterring others for reasons of security, immigration control, economic interest, and political interest [18]. Mau develops a related account of the global mobility divide, in which visa policies produce unequal access to cross-border movement [17]. In this literature, visa policy is not a neutral background condition. It is one of the central mechanisms through which access to international mobility is facilitated for some groups and constrained for others.

International student mobility is part of this wider mobility system. Oduwaye reviews the challenges faced by international students and shows that financial stability and other adaptation conditions remain central issues across host countries [19]. Gulel describes visa inequities in academic mobility and treats visa access as a form of privilege rather than a purely administrative step [9]. These studies support the premise that international education cannot be analysed only through enrolment demand or university admission. For many applicants, admission to a programme is only one step in a longer chain of documentary, financial, institutional, and immigration requirements.

The New Zealand case is relevant because international education is a policy and economic sector, while student entry remains governed by formal visa requirements. Craymer reports New Zealand's policy aim to expand the international education market substantially by 2034 [7]. At the same time, Spoonley discusses contemporary immigration policy challenges in New Zealand, and Bonnett reports concerns about Immigration New Zealand's processing systems and risk capacity [24, 5]. INZ's published student visa instructions require applicants to provide evidence and documents, including enrolment, tuition arrangements, sufficient funds, bona fide temporary-entry intent, and relevant health and character information [15]. These formal requirements make the student visa decision process a suitable empirical setting for studying how country-level conditions may appear in aggregate approval-rate patterns.

Existing quantitative migration research provides methodological precedents for joining migration-related outcomes with country-level indicators. Weber examines the relationship between development and international student migration, showing that student mobility can be studied as part of broader development and migration transitions [27]. Bertoli uses international migration data

joined with macroeconomic sources and highlights the importance of modelling migration in relation to multiple country-level contexts [2]. Abbott applies ordinary least squares regression in a gravity-model framework for international student migration and uses country-level variables to explain student flows [1]. These studies differ from the present analysis in outcome and design, but they support the use of country-level explanatory variables and regression-based approximation for migration-related questions.

Prior research also shows that visa policy and approval patterns can shape international student mobility. Chen studies U.S. F-1 visa restrictiveness and international student enrolment, while Gonnot examines Canadian student visa policy and international student migration [6, 8]. Gonnot is especially relevant because it links lower study-visa approval rates for applicants from developing countries to information and documentation problems that may fail to resolve immigration officers' concerns [8]. This supports the present article's cautious interpretation: lower approval rates need not be read as evidence of unequal person-to-person treatment; they may reflect differences in the evidence available to satisfy formal visa requirements.

Country-level and regional explanations are also supported by the broader student-mobility literature. Perkins and Neumayer, Abbott, Bessey, and Lanati and Thiele use country-level data to analyse determinants of international student flows, including geographic factors, networks, institutional quality, political freedom, GDP per capita, fees, and foreign aid projects [21, 1, 3, 16]. This literature justifies treating student mobility and visa outcomes as phenomena that can be associated with country-level conditions, while also showing that such models cannot identify all applicant-level mechanisms.

Some related research examines unequal outcomes in labour-market or social settings. Hersch studies the relationship between skin colour and immigrant labour-market outcomes in the United States, and Oreopoulos uses a field experiment to examine discrimination against applicants with foreign experience or non-English names [10, 20]. These studies are not direct models of INZ student visa decisions and are not used here to infer bias in visa assessment. Their relevance is narrower: they show that observed outcomes involving migrants can vary by origin-linked characteristics, and that empirical work must be careful to distinguish documented unequal treatment from aggregate differences that may arise through other mechanisms.

The present article contributes to this literature by analysing publicly available INZ fee-paying student visa decision data at country level and joining it with publicly available World Bank, WHO, and UN M49 reference data. Previous studies have used visa restrictiveness, student mobility, enrolment, or country-level development variables, but peer-reviewed work directly modelling country-level student visa approval rates with both development and health indicators appears limited. The analysis therefore addresses a specific empirical gap: whether country-level approval-rate variation in New Zealand student visa decisions is associated with application volume, national development indicators, income classification, and alternative regional classifications.

3 Data

The primary source data are publicly available Immigration New Zealand off-shore or overseas student visa decision tables for 2022, 2023, 2024, and 2025 [11–14]. The files are stored in `data/original/` and are not edited manually. They report fee-paying student visa decisions by applicant nationality or country. The yearly files use slightly different column names, but the common information is the country or nationality, number approved, number declined, total or volume of applications, approval rate, and decline rate.

The unit of observation in the prepared INZ table is a country or applicant nationality aggregated across the four-year period. The unified prepared file, `data/prepared/inz_student_visa_decisions_2022-2025.csv`, contains 179 countries or applicant nationalities. For each row it reports total approvals, total declines, total applications, approval rate, and `Applications_log10`, the base-10 logarithm of total applications. Approval rate is calculated as approved applications divided by total applications. This table is used for descriptive results on application volume and overall approval rates.

The modelling dataset joins the prepared INZ table to publicly available country-level reference data from the World Bank, the World Health Organization, and UN M49 [29, 28, 30, 31, 26]. The World Bank reference table contributes FY2026 region, income group, lending category, GDP per capita, GNI per capita using the Atlas method, life expectancy, internet-use percentage, and urban-population percentage. The WHO reference table contributes WHO region, under-five mortality, maternal mortality, DTP3 immunization coverage, and physician density. The UN M49 reference table contributes an alternative regional classification.

Country names are harmonised across the INZ, World Bank, WHO, and UN M49 files before modelling. The combined modelling file is `data/prepared/inz_2022-2025_wb_who_m49_combi`. It contains 110 countries after country matching, removal of rows with missing numerical values needed for the models, and exclusion of countries with fewer than 10 total applications across 2022–2025. Countries present in the prepared INZ table but absent from the regression table are reported separately in `data/support/Countries_excluded_from_regression.csv`.

The repository distinguishes source, prepared, combined, and result files. Source files in `data/original/` preserve the downloaded INZ and reference tables. Prepared files in `data/prepared/` contain harmonised and joined datasets used for analysis. Result files in `data/results/` contain model coefficients, scores, predictions, prediction-rate tables, and correlation outputs. Support files in `data/support/` contain derived diagnostic summaries and auxiliary tables used to interpret the results.

4 Methods

TODO: Describe the KNIME and Python workflow. Explain the unification of yearly INZ files, aggregation by country, calculation of approval rate and log10

applications, joins with reference data, correlation analysis, and linear regression models.

TODO: Explain the three regional model specifications: UN M49 regions, WHO regions, and World Bank regions. State that the models are descriptive or explanatory approximations, not causal designs and not operational prediction systems.

The country-level numerical variables and the regional variables have different methodological status. Health, income, internet-use, mortality, and life-expectancy measures are treated as country-level reference indicators. For a given source and year, these variables have a defined value for a country. Regional classification is different. A region is an external grouping imposed on countries, and the same country can be assigned to different regional systems depending on the classification scheme. For this reason, regional division is treated as a modelling choice rather than as an inherent property of the country.

The analysis therefore estimates four related experiments. Three models use alternative external regional classifications: UN M49 regions, World Bank regions, and WHO regions. A fourth model excludes regional variables. This design allows the analysis to separate two questions: whether country-level reference indicators are associated with approval rates, and whether the choice of regional grouping changes the fitted model.

The analysis uses several descriptive statistics, test statistics, and model diagnostics. Associations between approval rate and individual country-level variables are examined with Pearson correlation coefficients. These correlations are reported with the correlation coefficient, p-value, and degrees of freedom where relevant. This approach is used both for the relationship between application volume and approval rate and for relationships between approval rate and country-level indicators such as mortality, life expectancy, internet use, physician density, GNI per capita, and GDP per capita.

The interpretation of these statistics rests on INZ's published student visa instructions, which define common formal requirements for applicants. The models therefore do not use country-level differences to infer unequal treatment, bias, or discriminatory decision-making. They use those differences to study variation in the conditions under which applicants from different countries prepare, document, finance, and verify applications within the same formal decision framework.

The regression models are evaluated with standard linear model fit statistics. The main comparison across model specifications uses R-squared, mean absolute error, mean squared error, and root mean squared error. These statistics are used to compare the UN M49, World Bank regional, WHO regional, and no-region specifications. They measure fit within the current country-level modelling table and should not be read as validation of a causal mechanism.

Individual regression terms are assessed with coefficient estimates and p-values. The p-values are used to identify variables and regional indicators that cross the conventional $p < 0.05$ threshold within each specification. Because the models include correlated country-level indicators and categorical regional

variables, statistical significance is interpreted as conditional association within the fitted model rather than as evidence of an independent causal effect.

Prediction diagnostics are based on residuals and relative fitted error. Residual summaries include the mean residual, residual standard deviation, mean absolute residual, and maximum absolute residual. Residuals are also inspected against fitted values to check whether unexplained error is concentrated in particular fitted-value ranges. Relative residuals are examined with `Prediction_rate`, defined as the ratio of observed approval rate to fitted approval rate. Countries with observed approval rates below 80 percent or above 120 percent of the fitted value are treated as large proportional deviations for descriptive comparison.

Robustness checks compare the same quantities across alternative regional classifications and a weighted UN M49 specification. The weighted model uses `Applications_log10` as the observation weight, giving more influence to countries with larger log-transformed application volumes. It is evaluated with weighted R-squared, weighted mean absolute error, weighted mean squared error, and weighted root mean squared error, and also with unweighted fit statistics for comparison with the main UN M49 model.

5 Results

5.1 Descriptive Results

The unified INZ decision table contains 179 countries or applicant nationalities and 148,746 total fee-paying student visa applications across 2022–2025. Across these records, the overall approval rate is approximately 81.3 percent.

Application volume is unevenly distributed across countries. The largest application counts are from China, India, Japan, Nepal, the United States of America, Sri Lanka, Germany, Thailand, the Philippines, and South Korea. These high-volume countries do not all have the same approval rate. For example, China, Japan, the United States of America, Germany, and South Korea have approval rates above 96 percent, while India and Nepal have lower approval rates in the current prepared table.

The relationship between application volume and approval rate is positive but not strong. In the unified country-level table, the Pearson correlation between `log10` total applications and approval rate is about 0.27, with $p = 0.000254$ and 177 degrees of freedom. This supports a limited descriptive statement: countries with more applications tend to have higher approval rates, but application volume alone does not explain most of the variation in approval rates.

(Image of Total Application `log10` to Approval rate)

5.2 Correlation With Country-Level Indicators

After joining the INZ data with WHO, World Bank, and UN M49 reference data, the combined modelling table contains 110 countries. This reduction reflects the availability of matched country-level reference variables, the removal of rows

with missing numerical values, and the exclusion of countries with fewer than 10 total applications across the four-year period.

Several country-level indicators are strongly correlated with approval rate. Approval rate is negatively correlated with WHO under-five mortality per 1,000 live births ($r = -0.746$, $p = 1.24e - 30$) and WHO maternal mortality per 100,000 live births ($r = -0.681$, $p = 7.97e - 24$). Approval rate is positively correlated with life expectancy at birth ($r = 0.715$, $p = 4.19e - 27$), internet users as a percentage of population ($r = 0.633$, $p = 7.62e - 20$), physician density per 10,000 population ($r = 0.552$, $p = 1.56e - 14$), GNI per capita using the Atlas method ($r = 0.520$, $p = 8.18e - 13$), and GDP per capita in current USD ($r = 0.501$, $p = 7.42e - 12$).

The same correlation output also shows that several explanatory variables are strongly correlated with each other. For example, WHO under-five mortality is strongly negatively correlated with life expectancy ($r = -0.888$) and internet-user percentage ($r = -0.770$), and strongly positively correlated with WHO maternal mortality ($r = 0.881$). This means that the numerical country-level indicators partly measure overlapping dimensions of national development, health conditions, and institutional capacity.

These correlations indicate that approval rates are associated with broad country-level development, health, and income measures. They should not be read as causal effects. The data are aggregated by country, and the variables describe country-level conditions rather than individual applicants.

This interpretation is limited to formal and administrative conditions surrounding applications. The analysis is explicitly based on INZ's published visa instructions, which set out common formal requirements for student visa applicants, including evidence of enrolment, tuition arrangements, sufficient funds, bona fide temporary-entry intent, and relevant health and character requirements. The observed country-level differences are therefore not interpreted as evidence of officer behaviour, discriminatory intent, bias, or unequal treatment in individual application assessment. Rather, they are treated as aggregate descriptive outcomes that may reflect country-level differences in documentary, financial, institutional, and verification conditions under the same formal decision framework.

5.3 Regression Approximation

Four linear regression specifications were estimated. Three specifications use different regional classifications: UN M49, World Bank regions, and WHO regions. A fourth specification excludes regional classification variables. In each case, approval rate is the dependent variable, and the model uses selected WHO and World Bank country-level variables, with or without regional classification variables, as explanatory inputs.

The UN M49 specification has the strongest fit among the current models. Its R-squared is 0.867, with mean absolute error of 0.069 and root mean squared error of 0.090. The World Bank regional specification has R-squared of 0.812, mean absolute error of 0.082, and root mean squared error of 0.107. The WHO

regional specification has R-squared of 0.786, mean absolute error of 0.087, and root mean squared error of 0.114. The no-region specification has the weakest fit, with R-squared of 0.706, mean absolute error of 0.100, and root mean squared error of 0.134.

These scores suggest that the current country-level variables explain a substantial share of the variation in country-level approval rates, and that regional classification adds explanatory structure to the model. The no-region model still fits a large part of the variation, but all three regional specifications improve fit. The better fit of the UN M49 model may reflect the more detailed regional division used in that specification. This interpretation is supported by the robustness and sensitivity checks described below, but it should remain limited to model fit rather than causal explanation.

The coefficient tables show that model fit is not produced by regional variables alone. Other variables from the combined dataset also contribute to the fitted approval rates. The signs and magnitudes of these coefficients should be interpreted carefully because the model includes correlated country-level indicators and categorical variables. The coefficients show conditional associations within the fitted model, not independent causal effects.

In the UN M49 experiment, the non-region variables with p-values below 0.05 are urban population percentage (coefficient = -0.279, $p = 0.0053$), life expectancy at birth (coefficient = 0.499, $p = 0.0128$), and the missing category for World Bank lending category (coefficient = 0.242, $p = 0.0156$). Several UN M49 regional indicators are also below 0.05, including Northern Africa (coefficient = -0.254, $p = 0.0109$), Middle Africa (coefficient = -0.275, $p = 0.0219$), and Southern Asia (coefficient = -0.210, $p = 0.0336$). In this specification, life expectancy remains statistically visible after regional controls are included, while most other numerical country-level indicators do not.

The missing category for World Bank lending category needs a specific interpretation. In the modelling table, all countries in this category are high-income countries. They include countries such as Japan, Singapore, South Korea, Canada, the United States of America, Germany, France, Great Britain, Switzerland, Norway, Sweden, Austria, Belgium, Denmark, Ireland, Italy, Spain, Kuwait, Oman, Saudi Arabia, and the United Arab Emirates. The category therefore should not be read as ordinary data loss. In this dataset, it mainly identifies high-income economies without an assigned World Bank lending category. Its positive coefficient is consistent with the generally high approval rates observed for this group, but it should be interpreted as a proxy for this country group rather than as an effect of missing information.

In the World Bank regional experiment, the significant non-region variables are World Bank low-income status (coefficient = -0.407, $p = 5.48e - 05$), lower-middle-income status (coefficient = -0.136, $p = 0.0461$), the missing category for World Bank lending category (coefficient = 0.224, $p = 0.0171$), and urban population percentage (coefficient = -0.255, $p = 0.0061$). Significant World Bank regional indicators include Middle East, North Africa, Afghanistan and Pakistan (coefficient = -0.270, $p = 8.19e - 08$), Europe and Central Asia (coefficient =

-0.185, $p = 0.000369$), South Asia (coefficient = -0.232, $p = 0.000555$), and Sub-Saharan Africa (coefficient = -0.156, $p = 0.0042$). This model therefore combines income-group effects, urban population percentage, and broad World Bank regional effects.

In the WHO regional experiment, the significant non-region variables are World Bank low-income status (coefficient = -0.442, $p = 4.23e - 05$), lower-middle-income status (coefficient = -0.178, $p = 0.0127$), the missing category for World Bank lending category (coefficient = 0.231, $p = 0.0198$), and urban population percentage (coefficient = -0.237, $p = 0.0132$). Among WHO regions, the Western Pacific Region is significant (coefficient = 0.198, $p = 0.000975$). The Region of the Americas is close to the 0.05 threshold but does not cross it ($p = 0.0691$). Compared with the World Bank regional model, the WHO regional model has fewer significant region terms and weaker overall fit.

In the no-region experiment, the model has no regional indicators, so the fit depends only on non-region country-level variables. The significant variables are World Bank low-income status (coefficient = -0.609, $p = 2.50e - 07$), lower-middle-income status (coefficient = -0.224, $p = 0.00465$), WHO DTP3 immunization coverage (coefficient = -0.204, $p = 0.0152$), and urban population percentage (coefficient = -0.264, $p = 0.0081$). This confirms that the non-region variables carry substantial explanatory information, even though the no-region model fits worse than all three regional specifications.

(Table with coefficients)

(Images of Predicted Approval rate to Approval Rate for different regions division)

5.4 Prediction Deviations

The prediction tables compare observed approval rates with fitted approval rates. In all four specifications, the mean observed approval rate and mean fitted approval rate are both approximately 0.754 in the modelling table. Most fitted values are close enough to observed values to support the conclusion that the models capture major structure in the data.

(Reference to the table with excluded countries in Appendix)

The number of large proportional deviations differs by specification. Using the rule that a country is outside the expected band when the observed approval rate is below 80 percent or above 120 percent of the fitted value, the UN M49 model has 15 such countries, the World Bank regional model has 19, the WHO regional model has 25, and the no-region model has 28. This again suggests that the UN M49 model gives the closest fit among the current specifications.

In the UN M49 model, countries where the observed approval rate is lower than 80 percent of the fitted value include Chad, Afghanistan, Uzbekistan, Turkey, Myanmar, Cameroon, Yemen, Tanzania, Algeria, and Ghana. Countries where the observed approval rate is higher than 120 percent of the fitted value include Kazakhstan, Timor Leste, Nigeria, Sudan, and the Democratic Republic of Congo.

(Table for these countries)

These cases should not automatically be treated as technical outliers. They show where the current country-level variables and regional classifications do not fit the observed approval rate well. They may point to omitted variables, data quality issues, country-specific applicant composition, policy or processing factors, or other mechanisms not represented in the current dataset.

6 Robustness or Sensitivity Checks

The first sensitivity check is already built into the current workflow: the same modelling task is estimated with three alternative regional classifications and with no regional classification. The UN M49 model has the strongest fit, followed by the World Bank regional model, the WHO regional model, and the no-region model. This comparison supports the statement that regional classification matters for model fit and that the more detailed UN M49 classification fits the current data best.

The score tables support the same ordering across several metrics. R-squared declines from 0.867 in the UN M49 model to 0.812 in the World Bank regional model, 0.786 in the WHO regional model, and 0.706 in the no-region model. Mean absolute error and root mean squared error move in the opposite direction: they are smallest in the UN M49 model and largest in the no-region model.

The prediction-rate tables provide a second check. The UN M49 model produces the fewest countries outside the 80–120 percent fitted-value band: 15 countries, compared with 19 in the World Bank regional model, 25 in the WHO regional model, and 28 in the no-region model. This supports the view that regional variables improve the fit and that UN M49 is the strongest of the current regional specifications.

(Table with metrics and number of outliers)

The coefficient tables for the UN M49, WHO, World Bank, and no-region specifications are useful for comparing model outputs across alternative specifications. They show whether signs, magnitudes, and statistical significance change across region definitions. However, they do not test all modelling assumptions by themselves.

The coefficient stability summary in `data/support/Coefficients_stability_summary.csv` (add this table to Appendix) answers whether the same country-level variables remain important across specifications. It shows two patterns. First, urban population percentage is significant in all four specifications, with negative coefficients and p-values below 0.05. Second, World Bank income-group variables are especially important when UN M49 regions are not used: low-income status and lower-middle-income status are significant in the World Bank, WHO, and no-region models. In the UN M49 model, low-income status is weaker and does not reach the 0.05 threshold ($p = 0.0714$), suggesting that some income-related structure may be absorbed by the more detailed UN M49 regional categories.

The numerical health and development indicators are less stable across specifications. Life expectancy is significant in the UN M49 model ($p = 0.0128$) but

not in the World Bank, WHO, or no-region models. WHO DTP3 immunization coverage is significant only in the no-region model ($p = 0.0152$). GDP per capita, GNI per capita, physician density, under-five mortality, maternal mortality, and internet-user percentage do not remain statistically significant in the multivariable coefficient tables, even though several of them have strong bivariate correlations with approval rate. This difference is consistent with overlapping information among country-level development indicators.

This does not mean that these parameters are substantively unimportant. Their bivariate correlations show that they are part of the broader country-level pattern associated with approval rates. In the multivariable models, however, several indicators describe related dimensions of national income, health systems, infrastructure, and institutional development. Once these variables are entered together with income groups and regional classifications, their separate coefficients become harder to distinguish statistically. Their substantive importance is therefore better understood as part of a shared country-level development profile, rather than as isolated independent predictors.

The residual summaries and residuals-versus-fitted plots in `data/support/` provide an additional model check. Across all four specifications, mean residuals are approximately zero, which is expected for these fitted linear models. The residual spread follows the same ordering as the score tables: the UN M49 model has the smallest residual standard deviation (0.090) and mean absolute residual (0.069), followed by the World Bank regional model (0.108 and 0.082), the WHO regional model (0.115 and 0.087), and the no-region model (0.134 and 0.100). The largest absolute residual is also smallest in the UN M49 model (0.294) and largest in the no-region model (0.437). This supports the same conclusion as the score and prediction-rate checks: the UN M49 specification leaves less unexplained error than the alternatives.

(Table with these metrics)

The largest positive and negative relative residuals are reported in `data/support/Relative_residuals_pos` and `data/support/Relative_residuals_negative_by_model.csv`. These tables use `Prediction_rate`, the ratio of observed approval rate to fitted approval rate, rather than only the absolute residual. This is useful because the same absolute difference can be more consequential when the fitted approval rate is low. The positive table identifies countries where observed approval is high relative to the fitted value; the negative table identifies countries where observed approval is low relative to the fitted value.

The relative-residual tables also show that several countries recur among the largest positive or negative deviations across model specifications. This supports two interpretations. First, the model comparison is stable enough to support the conclusion that the UN M49 specification has the best fit among the current alternatives. Second, the largest country-level deviations are not explained only by the choice of regional classification. Persistent deviations across specifications may indicate omitted country-specific conditions, applicant-composition differences, documentation or verification factors, or other application-level mechanisms not represented in the current country-level dataset.

An additional weighted UN M49 experiment was estimated in Python using `Applications_log10` as the observation weight. This gives more influence to countries with larger log-transformed application volumes while preserving the same UN M49 feature structure. The weighted model remains close to the unweighted UN M49 model. Its weighted R-squared is 0.855, compared with the unweighted UN M49 R-squared of 0.867. Its weighted mean absolute error is 0.068 and weighted root mean squared error is 0.088, both close to the unweighted UN M49 values of 0.069 and 0.090. When evaluated without weights, the weighted model has R-squared of 0.864, mean absolute error of 0.071, and root mean squared error of 0.091. These results indicate that the main UN M49 fit result is stable when countries with larger log-transformed application volumes receive more weight.

The weighted coefficient table also supports a cautious interpretation. Life expectancy at birth and urban population percentage remain statistically significant in the weighted UN M49 model, with p-values of 0.0078 and 0.0051. Other variables that were significant in the unweighted UN M49 model, including several regional indicators and the missing World Bank lending-category term, no longer cross the 0.05 threshold. This pattern is substantively plausible. Weighting by application volume reduces the statistical prominence of several regional terms, suggesting that some region-level coefficient patterns in the unweighted model are influenced by treating each country equally regardless of application volume. When countries with more applications receive more weight, the overall UN M49 model fit remains similar, but the evidence for specific regional coefficients becomes weaker. The weighted experiment therefore strengthens the model-fit conclusion while reinforcing caution about individual coefficients.

These checks do not turn the analysis into a causal design. Their role is narrower: to show whether the descriptive model fit and the comparison between regional classifications are stable under reasonable alternative specifications.

7 Discussion

The results show that country-level variation in INZ fee-paying student visa approval rates is structured rather than random. Application volume has a positive but limited association with approval rate. This may reflect application ecosystems that develop around high-volume countries: education agents, advisers, institutional experience, previous successful applicants, and better knowledge of documentary expectations. The present data do not measure these mechanisms directly, so this explanation should be treated as an interpretation rather than a tested cause.

The joined WHO, World Bank, and UN M49 data show that approval rates are also associated with broad country-level conditions. Countries with higher life expectancy, higher internet-use rates, higher physician density, and higher income measures tend to have higher approval rates, while countries with higher mortality indicators tend to have lower approval rates. These variables should not be read as direct determinants of individual decisions. They are better un-

derstood as indicators of the administrative, economic, health, and institutional context from which applicants prepare evidence for a formal visa process.

The regression results support the same general interpretation. The no-region model already explains a substantial share of the variation, which means that non-region country-level variables contain important information. Adding regional classification improves model fit, and the UN M49 specification performs best among the current alternatives. This suggests that regional structure matters, but it does not mean that region itself is a causal factor. Regional classifications are summary categories that may capture shared institutional, economic, mobility, documentation, and applicant-composition conditions that are not directly observed in the dataset.

The weighted UN M49 experiment adds an important qualification. When countries with larger log-transformed application volumes receive more weight, the overall model fit remains close to the unweighted UN M49 model, but several regional terms and the missing World Bank lending-category term no longer cross the $p < 0.05$ threshold. This supports the main fit conclusion while reducing confidence in strong interpretations of individual regional coefficients. Some regional patterns are more visible when each country is treated equally, and weaker when high-volume applicant countries receive more influence.

The residual and relative-residual checks show where the current model does not explain observed approval rates well. Positive deviations, where observed approval is higher than fitted approval, may reflect country-specific conditions not captured by the model. Possible examples include intergovernmental or inter-educational relationships, established pathways between education providers, scholarship or sponsorship channels, or unusually effective application support structures. These possibilities are not measured in the current data, but they identify plausible directions for further research.

Negative deviations require even more caution. Countries with observed approval rates well below fitted values cannot be explained by the current country-level predictors alone. These cases may involve applicant-composition differences, programme choice, documentation problems, financial-evidence issues, verification difficulties, or other application-level factors that are absent from the dataset. They should therefore be treated as questions for further empirical investigation, not as evidence for a specific institutional mechanism.

Overall, the analysis supports a narrow conclusion: country-level approval-rate differences are associated with application volume, national development indicators, income categories, and regional classification systems. Within the interpretive framework of this article, these differences are treated as possible evidence of different applicant conditions across countries and regions under the same formal decision framework. The results are useful for describing and modelling variation at the country level, but they do not identify the individual-level reasons why an application is approved or declined.

8 Limitations

The analysis is limited by aggregation at the country level. The unit of observation is country or applicant nationality, not an individual application. The models therefore cannot separate applicant-level characteristics, institutional documentation quality, programme choice, education provider characteristics, or officer-level assessment details.

The country-level predictors also overlap substantially. The correlation table shows strong pairwise correlations among health, income, internet-use, mortality, and life-expectancy indicators. This supports the substantive interpretation that these variables describe a shared country-level development profile, but it limits interpretation of individual regression coefficients. A full variance inflation factor analysis or other design-matrix diagnostic could examine this more formally, but the present article treats the existing correlation output as sufficient evidence that individual coefficients should be interpreted cautiously.

The results are also sensitive to country matching, missing numerical values, regional classification choices, and the exclusion of countries with fewer than 10 total applications across 2022–2025. The comparison across UN M49, World Bank, WHO, and no-region specifications addresses part of this sensitivity, but it does not remove the broader limitations of observational, aggregated data.

9 Conclusion

This article shows that country-level variation in Immigration New Zealand fee-paying student visa approval rates during 2022–2025 is associated with application volume, country-level development indicators, income categories, and regional classification systems. The strongest current model fit is obtained with the UN M49 regional specification, but the results also show that non-region country-level variables carry substantial explanatory information. The analysis should therefore be read as a country-level descriptive and modelling exercise, not as a causal account of individual visa decisions or as evidence about officer-level assessment behaviour.

The main empirical contribution is to connect INZ student visa decision outcomes with external country-level reference data in a reproducible workflow. This makes visible both broad structure and unexplained country-specific deviations. The broad structure supports cautious interpretation of country-level application conditions. The deviations identify where the current data are insufficient and where more detailed research is needed.

It is difficult to assume that WHO and World Bank indicators directly influence visa approval rates. A more cautious interpretation is that these indicators mark broader country-specific conditions that are not directly observed in the present dataset. Such hidden conditions may influence both the measured WHO and World Bank variables and the practical circumstances under which applicants prepare visa evidence. Identifying these country-specific mechanisms is therefore a central task for further research.

Future work should extend the comparison beyond New Zealand by examining student visa decisions in other receiving countries. It should also add further country-level variables from reliable sources, including measures related to crime, religion, culture, institutional trust, and other social conditions where the definitions and provenance are clear. A further direction is to examine whether inter-country treaties, education-sector relationships, diplomatic communication, scholarship channels, or other formal links between countries help explain why some countries have approval rates above or below the values fitted by the current model.

To inform our empirical analysis, we propose a basic theoretical model to understand how incomplete information affects the processing of visa applications by immigration officers.

10 Appendices or Supplementary Material

TODO: Add workflow details, variable definitions, joined-data diagnostics, model diagnostics, and additional result tables.

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